Enrique Amaya

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Education

Embry-Riddle Aeronautical University — Daytona Beach, FL
Accelerated Master's Program: Mechanical Engineering — December 2028
Area of Concentration: Robotics & Autonomous Systems

Embry-Riddle Aeronautical University — Daytona Beach, FL
Bachelor of Science: Engineering Physics — May 2027
Area of Concentration: Spacecraft Systems — GPA: 3.4/4.0
Minors: Astronomy & Astrophysics, Applied Mathematics

Work Experience

Undergraduate Researcher — 08/2025 – Present
Embry-Riddle Aeronautical University – Daytona Beach, Florida
- Conducting undergraduate research at the Engineering Physics Propulsion Laboratory (EPPL), engaging in projects ranging from robotics, autonomous systems, and electrical/mechanical engineering.
- Handled and organized important laboratory equipment such as electronics, hardware, and robotic components.

Academic Advancement Center Physics Tutor — 01/2025 - Present
Embry-Riddle Aeronautical University – Daytona Beach, Florida
- Tutoring in the AAC physics lab aiding undergraduate students with fundamental concepts from physics I, II & III courses.
- Assisted the Chair of the Physics Department in Recitations, an assigned time where students receive specific physics help.

Project Experience

Autonomous Intelligent Real-Time Hunting of Uncrewed Navigating Drones — 08/2025 - Present
Engineering Physics Propulsion Laboratory (EPPL) – Embry-Riddle Aeronautical University
- Led AIRHOUND's tracking & geometry section by constructing camera-based target localization methods.
- Applied camera pinhole mathematics to convert 2D bounding box detections into angular yaw correction commands.
- Implemented a 3D Kalman filter for target position and velocity estimations and predictions in situations of target dropouts.
- Collaborated to integrate Kalman-state, PINN prediction and yaw-command outputs through Jetson ROS 2 pipeline.

BalloonSat Payload Design: Sunlight Tracking System — 03/2026 – 05/2026
Microcomputers & Electronic Instrumentation Laboratory – Embry-Riddle Aeronautical University
- Designed a multifunctional sun tracking system capable of monitoring environmental and system-level parameters intending to replicate real payload conditions in actual space satellite missions.
- Constructed a photodiode and stepper motor-based tracking subsystem to orient payload toward a light source.
- Developed non-inverting amplifier circuits in order to use thermistors to measure the temperatures across payload.
- Built two voltage regulators in order to provide both positive and negative power supply terminals.
- Data was processed through a microcontroller and transmitted via UART to a MATLAB ground station in real-time.

Skills

- **Programming Software:** C, Python, C++, ROS2, Atmel Studio, GitHub, Bambu Studio.
- **Technical Skills:** Electronic Instrumentation, CATIA V5, NI Multisim 14.2, 3D Printing, Microsoft Office.
- **Languages:** English (Fluent), Spanish (Fluent), French (Beginner), Portuguese (Beginner).

Leadership

Student Athlete Advisory Committee (SAAC) Representative — 01/2025 – 05/2026
Embry-Riddle Aeronautical University – Daytona Beach, Florida
- Member of Embry-Riddle's NCAA Division II Men's Tennis Team and acted as the SAAC representative.
- Represented the Men's Tennis Team's interests towards Embry-Riddle's multiple university boards.
- Led in coordinating athletic events, fundraisers, and volunteering events.

Publications

[1] R. Malarchick, J. Castelblanco, E. Amaya, C. DiMario, and G. Brinkman, "Robust Real-Time Target Tracking with Onboard Vision-Based Yaw Control". *ERAU Beyond: Undergraduate Research Journal,* vol. 9, no. 1, Art. 6, Jun. 2026

[2] R. Malarchick, C. DiMario, E. Amaya, and G. Brinkman, "Predictive Target Pursuit for Autonomous UAVs using RF-DETR with Depth-Aware State Estimation and Physics-Informed Trajectory Prediction", *Proc. SPIE Defense + Commercial Sensing 2026,* vol. 14030, Jun. 2026.